

GURVIR SASAN

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EDUCATION

University of Toronto - HBSc, Computer Science

2019 - 2024

Minor - Mathematics & Statistics | Graduated with Distinction

EXPERIENCE

Software Engineer II

Apr 2025 – Present

Relay Financial

Toronto, ON

- Led end-to-end design and development of the event-driven **Call-Check** verification feature — a secure caller identity validation system, owning **architecture**, **system design**, and **TDD**; built with Webhooks, AWS SQS/Lambda, Redis, and React Native to validate **13,000+** monthly calls and reduce impersonation fraud.
- Implemented in-house **Magic Link authentication** leveraging JWT, Redis, AWS SES/Lambda and Castle SDK for fraud detection reducing account-takeover (**ATO**) historically totaling **~1.4M**.
- Architected and delivered a fine-grained, **policy-based authorization framework** replacing legacy **RBAC**; implemented a **facade-driven**, feature-flagged migration while refactoring edge-case logic and aligning backend APIs and frontend access flows to ensure backward compatibility and zero regressions.
- Implemented **Kafka** event consumption to migrate analytics and domain events off Segment, supporting an **event-driven microservices architecture** and enabling scalable backend workflows and frontend analytics.
- Revamped Team Management and Settings pages in **React**, improving performance and maintainability through component restructuring and **RTK Query**-based data fetching.

Software Developer

Oct 2024 – Apr 2025

Nojumi Solutions Inc.

Toronto, ON

- Designed and implemented a **multi-tenant, configuration-driven** full-stack feature enabling firms to define custom matter types and practice areas, increasing platform versatility by **40%**.
- Led a third-party integration with **Stewart Title**, designing secure **OAuth 2.0** authentication flows with CSRF protection for enterprise customers.
- Improved **UX** workflows and resolved frontend issues in **Angular** for **100+ clients** based on feedback, while delivering end-to-end solutions with **Scala** backend API development and database enhancements.

Software Developer - Practicum Project

Sept 2023 – Dec 2023

University of Toronto Mississauga

Mississauga, ON

- Led development of a permission-level **feature** on the university's TA application portal, enabling admins to edit student applications involving database modifications, new endpoints and frontend changes.

Junior Software Developer

May 2022 – Aug 2023

JANA Corporation

Aurora, ON

- Resolved **450+** frontend defects over a **16-month** internship, improving application stability and overall UX.
- Upgraded **React (16.x → 17.x)** and refactored legacy class-based components into hooks-based functional components, reducing unnecessary re-renders and API calls by **30%** and improving page load times by **23%**.
- Architected a secure **RESTful API** abstraction layer using **JWT authentication** and common design patterns, collaborating with backend and mobile teams to improve security and integration reliability.

PROJECTS

CallGPT - DeerHacks Hackathon Winner | *React, Flask, SQLite, Twilio, OpenAI*

2023

Restify | *React, Django, Django REST Framework, PostgreSQL, JWT*

2022

UofTBoard - Google Developer Student Club (GDSC) | *React, Node.js, Node, Express, Jest*

2021–2022

Image Recognition - Handwriting recognition machine learning | *Python, k-NN, Decision Trees*

2021

TECHNICAL SKILLS

Languages: TypeScript (ES6), Python, Scala, Java, Kotlin C, Bash, SQL, HTML/CSS

Technologies: React, React Native, Angular, Redux, Node.js, AWS (Lambda, API Gateway, SQS, SES, S3, Secrets Manager, CloudWatch), Express, Play Framework, Kafka, Redis, Webhooks, RTKQ, Datadog, PostgreSQL, Django, SQLite, ESRI, Docker, Git, Jest, Jira, Postman, Bruno